

LECTURE 1.2



Expectation and Conditional Means

ECONXXXXX Econometrics I

Sam Deegan
University College Dublin
PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

Last Week's Lecture

Lecture 1.1 set the module's question and the vocabulary for describing data:

1 How This Module Works.

What the term covers, how it is assessed, and the standing question.

2 Populations and Samples.

The question is about a population; the data is a sample.

3 Describing One Variable.

Centre, spread and shape, and what each one hides.

4 Describing Two Variables.

Covariance, correlation, and what one number cannot show.

Lecture Plan

1 **Expectation.**

A property of a distribution, not of a sample.

2 **The Algebra of Expectations.**

The rules every derivation in this module spends.

3 **The Conditional Expectation Function.**

The best guess of earnings given schooling.

SECTION 1

Expectation

1. Expectation
2. The Algebra of Expectations
3. The Conditional Expectation Function

Defining the Expectation of a Random Variable

A random variable is a quantity the draw decides. Its expectation says where the distribution sits, before any sample:

$$E[y] = \sum_k y_k p_k \quad (1)$$

- y : the variable, here hourly earnings in euro
- y_k : one value that y can take
- p_k : the share of the population sitting at that value
- $E[y]$: the population average of y , here 11.52 euro

Notation for the Population Mean and the Sample Mean

One average is a fixed fact; the other changes with every sample drawn.

The Population Mean Does Not Move

$E[y]$ is the population average: one fixed number, a property of the distribution. $\bar{y}$ is the sample average, 12.25 in our 120. Draw a different 120 and $\bar{y}$ is a different number.

The Population Mean Is Not Directly Observable

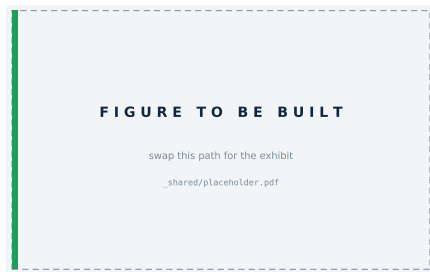


Figure 1: Population earnings in grey, the sample of 120 in colour, both means marked.

Only the Sample Mean Changes

Redraw the 120: the dashed line leaves 12.25, the solid stays at 11.52. $E[y]$ belongs to the population, not to the draw.

The Gap Between the Two Lines Has Its Own Distribution

Every claim this module makes is about the solid line. Every number we compute comes from the dashed one.

Lectures 1.3 and 2.3 Measure the Gap

The 0.73 between 12.25 and 11.52 is not a mistake to be removed. That gap changes with every sample drawn. A standard error puts a number on it.

SECTION 2

The Algebra of Expectations

1. Expectation
2. The Algebra of Expectations
3. The Conditional Expectation Function

The Three Rules of Expectation Algebra

Three rules give the average of a combination without knowing its distribution:

$$E[c] = c \qquad E[a + b y] = a + b E[y] \qquad E[y + z] = E[y] + E[z] \qquad (2)$$

- c, a, b : constants, fixed before any data arrive
- z : a second random variable

Additivity Needs No Independence

Averages add whatever the joint distribution is; y and z may be dependent.

Variance and Covariance Written as Expectations

Spread and co-movement are themselves averages, so the three rules reach them:

$$\text{Var}(y) = E[(y - E[y])^2] = E[y^2] - (E[y])^2 \quad (3)$$

$$\text{Cov}(y, x) = E[(y - E[y])(x - E[x])] = E[yx] - E[y]E[x] \quad (4)$$

- x : years of schooling
- $\text{Var}(\cdot)$: the variance of whatever sits inside
- $\text{Cov}(\cdot, \cdot)$: the covariance of the two inside

Why Expectations Do Not Pass Through a Product

An expectation passes through sums, constants and scalings, and not through a product. Independent means knowing one variable tells you nothing about the other. Here earnings rise 1.30 euro with each extra year of schooling. So schooling and earnings have a positive covariance.

The Product Rule Needs Zero Covariance

$E[yx]$ equals $E[y]E[x]$ exactly when $\text{Cov}(y, x)$ is zero. Independence delivers zero covariance, and the equality does not require independence.

The Variance of a Sum Carries a Covariance Term

Adding two random variables does not add their variances:

$$\text{Var}(y + z) = \text{Var}(y) + \text{Var}(z) + 2 \text{Cov}(y, z) \quad (5)$$

Zero Covariance Drops the Term

The covariance term drops out when y and z are uncorrelated. Independence is stronger than this rule asks for.

Where Each Rule Is Used Later in the Module

The rules are not conventions, and each one buys something later:

- **Linearity:** cuts the Lecture 2.3 proof of unbiasedness to two lines from twenty.
- **The Variance of a Sum:** builds the standard error in Lecture 2.3.
- **Covariance Over Variance:** is the slope Lecture 2.1 fits by least squares.

Each Rule Is the Sum Rearranged

Each rule is the definition of an expectation with the sum rearranged. The appendix writes the weighted sum, splits it, then takes the constants outside.

The Average of Schooling Times Earnings

Average schooling is about 13 years and average earnings about 11 euro.

Question

Is the average of (schooling $\times$ earnings) about 143?

SECTION 3

The Conditional Expectation Function

1. Expectation
2. The Algebra of Expectations
3. The Conditional Expectation Function

Defining the Conditional Expectation of Earnings Given Schooling

Split everyone by years of schooling, then average earnings inside each group:

$$E[y \mid x] \tag{6}$$

- $E[y \mid x]$: the average of y among people with that value of x

In a sample, that instruction is carried out with the mean from Lecture 1.1:

$$\hat{E}[y \mid x = 12] = \frac{1}{n_{12}} \sum_{i: x_i=12} y_i \tag{7}$$

- n_{12} : how many people in the sample have twelve years of schooling

A Function, Not a Number

$E[y \mid x]$ is the conditional expectation function, the CEF for short. The CEF returns a different average at each value of x .

Earnings Distributions at Three Levels of Schooling

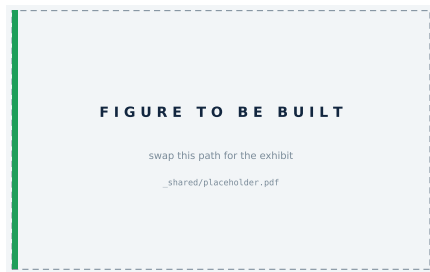


Figure 2: Earnings among people with ten, twelve and sixteen years of schooling.

Wide Spread Around Each Mean

At twelve years the conditional mean is 9.50. Earnings there scatter widely: the conditional standard deviation is 4.50.

The Conditional Expectation Function in This Population

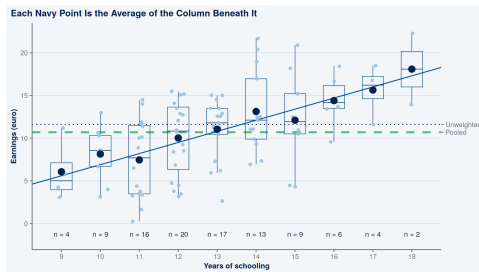


Figure 3: One hundred people at ten levels of schooling. Each navy point is the average of the column beneath it; the count in that column is written under it.

The Counts Are the Weights

Here $E[y | x] = 9.50 + 1.30(x - 12)$, the blue line. Pooling all one hundred gives 10.70. Averaging the ten column means without their counts gives 11.61, because it treats the two people at eighteen years as it treats the twenty at twelve.

The Sense in Which the Conditional Mean Is Best

No rule built on schooling beats the conditional mean on squared error:

$$E[y | x] = \arg \min_g E[(y - g(x))^2] \quad (8)$$

- $g(x)$: any function of x used to guess y
- $\arg \min_g$: the g that makes what follows as small as it goes

Both Expected Squares Must Be Finite

Both $E[y^2]$ and $E[g(x)^2]$ must be finite. Without them there is no expected squared error to compare.

Why the Minimum Sits at the Conditional Mean

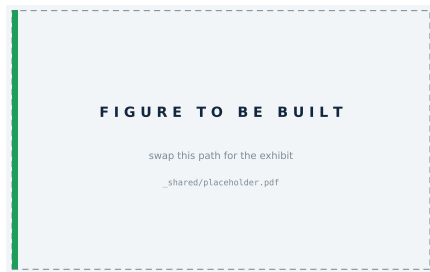


Figure 4: Expected squared error at twelve years of schooling, against the guess made.

The Floor Is the Conditional Variance

The curve bottoms out at 9.50, on a floor of 4.50^2 . That floor is the conditional variance, and no guess goes below it.

What the Fitted Line in Lecture 2.1 Approximates

Without the CEF, a fitted line is just the easiest curve to draw. Angrist & Pischke (2015) build the regression chapters of *Mastering Metrics* on the CEF.

The Line Approximates the CEF

Lecture 2.1 fits a line by least squares to approximate the CEF. This population's CEF is straight; real ones need not be. The CEF exists in the population before anyone fits anything.

Description Versus Causation in the Conditional Mean

Describing who earns what is not the same as saying what schooling does.

A Description, Not an Effect

$E[y | x]$ describes the population and says nothing about changing x . “People with sixteen years of schooling earn 14.70 on average” is a description. “Four more years raises earnings by 5.20” is a different claim. Nothing today supports it.

What Would Have To Be True?

This week the question is asked of the conditional mean of earnings.

Question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

Reading $E[y | x]$ as an effect is an assumption about who stayed in school. Stayers and leavers must differ in nothing else touching earnings.

Application

The CEF explorer cuts schooling into one-year bins. Each bin gets one point, at its mean of earnings.

Question

With our 120 people, will the bin means look like a line?

The lab repeats the exercise on all 200,000.

Summary: Expectation, Its Algebra, and the CEF

Today was three ideas, each doing one job:

1 **Expectation.**

A property of the distribution, not of the data you hold.

2 **The Algebra of Expectations.**

Three rules carry every derivation, and none of them covers a product.

3 **The Conditional Expectation Function.**

The CEF describes the population and does not intervene in it.

Before Lecture 1.3

1 **Play:**

The CEF explorer, widening the bin from one year to four.

2 **Apply:**

Problem set 1, covering 1.1 and 1.2, with both appendix derivations.

3 **Review:**

Wooldridge (2020), Math Refresher B-4e and B-4f, on conditional expectation.

Next Week

Lecture 1.3 computes an expectation from 120 people instead of a population:

- **Estimators:** rules that turn 120 numbers into one guess at $E[y]$.
- **Sampling Variation:** how far the guess lands from $E[y]$ across samples.

Thank you

sam.deegan@ucdconnect.ie
sam-deegan.com



BACKUP

Appendix

Derivation: The Algebra of Expectations

Write each expectation as its weighted sum, split it, take the constants outside.

Three Rules, the Variance Form Follows

Three rules follow: a constant, a linear function, and a sum. The variance form $E[y^2] - (E[y])^2$ is a consequence of them, not a fourth rule.

Derivation: The Conditional Mean Minimises Expected Squared Error

Add and subtract $E[y \mid x]$ inside the square, then expand. The cross term has expectation zero.

The Cross Term Has Mean Zero

Assume $E[y^2]$ and $E[g(x)^2]$ are finite. What remains is the conditional variance plus a squared term, zero only at $g(x) = E[y \mid x]$.

Notes and Tools

- **Binning in R:** `cut()` and `tapply()`, which is the whole binned-means lab.
- **The Lecture in One Line:** `mean(y[x == 12])` against `mean(y)`.
- **This Week's Script:** the CEF explorer, and the code that draws its exhibits.

References

Angrist, J. D., & Pischke, J.-S. (2015). *Mastering 'metrics: The path from cause to effect*. Princeton University Press.

Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage Learning.